A Flipped Unified Power Quality Conditioner for Active Distribution Networks: Topology, Modelling, and Control Strategy

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***Abstract*—The increasing penetration of distributed renewable energy resources (DERs) into medium-voltage (MV) and low-voltage (LV) distribution networks introduces new challenges for both power quality and grid stability. Conventional mitigation devices—dynamic voltage restorers (DVRs), static synchronous compensators (STATCOMs), and uninterruptible power supplies (UPS)—address these issues in isolation and adopt a purely reactive role. This paper proposes a *Flipped Unified Power Quality Conditioner* (F-UPQC), an architecture in which the shunt converter faces the grid to provide ancillary services to the distribution system operator (DSO), while the series converter faces the load to guarantee voltage stability for sensitive industrial loads. A shared DC link with optional energy storage completes the topology, enabling participation in local flexibility markets without additional hardware. The mathematical model of the complete system is derived, and a decoupled control strategy is presented. Theoretical analysis demonstrates the ability of the proposed topology to simultaneously compensate voltage sags and swells at the load terminals while injecting reactive power and mitigating current harmonics on the grid side. The proposed architecture is aligned with the evolving regulatory framework of the European Clean Energy Package and the growing demand for grid-edge devices capable of active distribution network (ADN) support.**

***Index Terms*—power quality, unified power quality conditioner, UPQC, active distribution network, ancillary services, dynamic voltage restorer, STATCOM, energy storage, voltage sag compensation, reactive power control**

I. Introduction

The decarbonisation of the European energy system is driving a rapid and large-scale deployment of distributed energy resources (DERs) at the medium-voltage (MV) and low-voltage (LV) distribution level. Photovoltaic plants, wind micro-turbines, battery storage systems, and electric vehicle chargers are transforming distribution grids from passive, unidirectional infrastructures into dynamic, bidirectional active distribution networks (ADNs) [5]. The European Commission has estimated investment requirements of 375–425 billion EUR in distribution infrastructure by 2030, an increase of 50–70 % compared to the previous decade [6].

This structural transformation carries two simultaneous and partially conflicting challenges. On one hand, the proliferation of power-electronics-interfaced generation and storage introduces harmonics, voltage unbalance, and rapid voltage fluctuations that degrade power quality for sensitive industrial loads [9]. On the other hand, the reduction of synchronous generation reduces system inertia and reactive power availability, placing new burdens on distribution system operators (DSOs) to procure ancillary services at the distribution level [7]. Industry studies estimate that power quality disturbances cost European and North American industries in excess of 150 billion USD annually in lost productivity and equipment damage [10].

Existing mitigation technologies address these challenges separately. Dynamic Voltage Restorers (DVRs) [2] inject a compensating series voltage to protect sensitive loads from sags and swells, but provide no service to the grid. Static Synchronous Compensators (STATCOMs) [3] and Active Power Filters (APFs) [4] operate on the shunt path and can provide reactive power support or harmonic mitigation, but offer no load-side protection. The Unified Power Quality Conditioner (UPQC), first proposed in [1], combines both functions by coupling a shunt and a series converter through a common DC link. However, in the classical UPQC configuration, the shunt converter is placed on the *load side* to compensate load current harmonics, while the series converter is placed on the *grid side* to compensate supply voltage disturbances. This arrangement prioritises the perspective of the end user and provides limited benefit to the DSO.

The present paper proposes a *topological inversion* of the standard UPQC architecture—hereafter referred to as the *Flipped UPQC* (F-UPQC). In the proposed configuration:

- the **shunt converter** is connected on the *grid side*, enabling provision of reactive power support, harmonic current injection, and fast voltage regulation as ancillary services to the DSO;
- the **series converter** is connected on the *load side*, ensuring full voltage stability for connected sensitive loads regardless of grid disturbances;
- an **energy storage** element on the shared DC bus decouples active power exchange between the two converters and enables participation in distribution-level flexibility markets.

The main contributions of this paper are:

1) A novel F-UPQC topology specifically designed for MV/LV distribution networks operating in the context of ADNs;

2) A complete small-signal mathematical model of the F-UPQC in the synchronous reference frame (dq frame);
3) A decoupled control architecture separating the series voltage-stabilisation loop from the shunt ancillary-service loop;
4) A steady-state power flow analysis demonstrating the feasibility of simultaneous load protection and DSO service provision;
5) A discussion of the regulatory context (EU Clean Energy Package, EN 50160, IEC 61000) and the positioning of the F-UPQC in emerging DSO flexibility markets.

The remainder of the paper is organised as follows. Section II describes the F-UPQC topology and compares it with the classical UPQC. Section III derives the mathematical model. Section IV presents the control strategy. Section V analyses the steady-state power flow and component ratings. Section VI discusses regulatory alignment. Section VII draws conclusions and outlines future work.

II. F-UPQC Topology

A. Classical UPQC vs. Flipped UPQC

Fig. 1 illustrates the structural difference between the classical UPQC and the proposed F-UPQC. In the classical topology, the shunt converter compensates non-linear load currents on the load bus, while the series converter mitigates supply-side disturbances. In the F-UPQC, the two converters exchange roles with respect to their point of connection: the shunt VSC interfaces with the grid at the point of common coupling (PCC), and the series VSC injects a compensating voltage on the load feeder through a series injection transformer.

[PLACEHOLDER: Classical UPQC vs. F-UPQC block diagram]

Figure 1: Structural comparison between (a) classical UPQC and (b) proposed Flipped UPQC (F-UPQC).

B. Component Description

The F-UPQC consists of the following main components:

1) Grid-side coupling transformer: A medium-voltage transformer with Dyn11 vector group provides galvanic isolation and a 30° phase shift that reduces low-order current harmonics reflected on the MV bus.

2) Shunt VSC (grid-side): A two-level or three-level voltage-source converter connected in shunt at the PCC via a coupling reactor L_{sh}. Operating as a grid-following controlled current source, it injects reactive current i_q on demand from the DSO and a harmonic cancellation current i_h computed from the grid current measurement.

3) Series VSC and injection transformer: A voltage-source converter whose AC terminals are connected to the primary of a series injection transformer T_{s}. The secondary winding is inserted in series on the load feeder. The converter synthesises a compensating voltage $\boldsymbol{v}_{\text{inj}}$ such that the load terminal voltage $\boldsymbol{v}_{\text{load}}$ tracks a clean sinusoidal reference $\boldsymbol{v}_{\text{ref}}$ independently of grid disturbances.

4) Common DC bus: A capacitor bank C_{dc} maintains the DC link voltage V_{dc} at a regulated level. The shunt VSC is responsible for DC bus voltage regulation in normal operation.

5) Energy storage interface: An optional bidirectional DC/DC converter connects an energy storage system (battery, supercapacitor, or hybrid) to the DC bus. During sustained voltage sags, the storage supplies the active power demanded by the series VSC, extending compensation capability beyond the steady-state limit dictated by grid power flow alone.

III. Mathematical Model

A. Reference Frame and Notation

All quantities are expressed in the synchronous dq reference frame rotating at the fundamental angular frequency $\omega_0 = 2\pi f_0$, with $f_0 = 50\,\text{Hz}$. The Park transformation adopted is power-invariant:

$$\boldsymbol{T}(\theta) = \sqrt{\frac{2}{3}} \begin{bmatrix} \cos\theta & \cos\!\left(\theta - \frac{2\pi}{3}\right) & \cos\!\left(\theta + \frac{2\pi}{3}\right) \\ -\sin\theta & -\sin\!\left(\theta - \frac{2\pi}{3}\right) & -\sin\!\left(\theta + \frac{2\pi}{3}\right) \end{bmatrix} \tag{1}$$

where $\theta = \omega_0 t + \theta_0$ is estimated by a synchronous-reference-frame phase-locked loop (SRF-PLL).

B. Shunt VSC Model

Let $\boldsymbol{v}_{\text{pcc}} = [v_{d,\text{pcc}},\, v_{q,\text{pcc}}]^\top$ be the PCC voltage and $\boldsymbol{i}_{\text{sh}} = [i_{d,\text{sh}},\, i_{q,\text{sh}}]^\top$ the shunt converter output current, both in the dq frame. The coupling reactor dynamics are:

$$L_{\text{sh}} \frac{d}{dt} \begin{bmatrix} i_{d,\text{sh}} \\ i_{q,\text{sh}} \end{bmatrix} = \begin{bmatrix} v_{d,\text{sh}} \\ v_{q,\text{sh}} \end{bmatrix} - \begin{bmatrix} v_{d,\text{pcc}} \\ v_{q,\text{pcc}} \end{bmatrix} - R_{\text{sh}} \begin{bmatrix} i_{d,\text{sh}} \\ i_{q,\text{sh}} \end{bmatrix} + \omega_0 L_{\text{sh}} \begin{bmatrix} i_{q,\text{sh}} \\ -i_{d,\text{sh}} \end{bmatrix} \tag{2}$$

where $\boldsymbol{v}_{\text{sh}}$ is the converter output voltage and R_{sh} represents switching and resistive losses.

C. Series VSC Model

Let n_s be the turns ratio of the series injection transformer and $\boldsymbol{v}_{\text{inj}}$ the injected voltage referred to the secondary. The load terminal voltage is:

$$\boldsymbol{v}_{\text{load}} = \boldsymbol{v}_{\text{pcc}} + \boldsymbol{v}_{\text{inj}} - Z_{\text{line}}\, \boldsymbol{i}_{\text{load}} \tag{3}$$

The series converter dynamics in the dq frame follow:

$$L_{\text{se}} \frac{d}{dt} \begin{bmatrix} i_{d,\text{se}} \\ i_{q,\text{se}} \end{bmatrix} = \begin{bmatrix} v_{d,\text{se}} \\ v_{q,\text{se}} \end{bmatrix} - n_s \begin{bmatrix} v_{d,\text{inj}} \\ v_{q,\text{inj}} \end{bmatrix} - R_{\text{se}} \begin{bmatrix} i_{d,\text{se}} \\ i_{q,\text{se}} \end{bmatrix} + \omega_0 L_{\text{se}} \begin{bmatrix} i_{q,\text{se}} \\ -i_{d,\text{se}} \end{bmatrix} \tag{4}$$

D. DC Link Model

The DC bus energy balance is:

$$C_{\text{dc}} \frac{dV_{\text{dc}}}{dt} = i_{\text{sh,dc}} - i_{\text{se,dc}} + i_{\text{sto}} \tag{5}$$

where $i_{\text{sh,dc}}$ and $i_{\text{se,dc}}$ are the DC currents drawn by the shunt and series VSCs respectively, and i_{sto} is the storage current (positive when discharging). In the power-invariant dq frame:

$$i_{\text{sh,dc}} = \frac{v_{d,\text{sh}}\, i_{d,\text{sh}} + v_{q,\text{sh}}\, i_{q,\text{sh}}}{V_{\text{dc}}} \tag{6}$$

$$i_{\text{se,dc}} = \frac{v_{d,\text{se}}\, i_{d,\text{se}} + v_{q,\text{se}}\, i_{q,\text{se}}}{V_{\text{dc}}} \tag{7}$$

IV. Control Strategy

A. Overview

The F-UPQC control is decomposed into four decoupled loops, as shown in Fig. 2:

1) **Shunt current controller** — inner PI loop regulating $i_{d,\text{sh}}$ and $i_{q,\text{sh}}$ with cross-coupling feed-forward decoupling;
2) **DC bus voltage controller** — outer PI loop providing the d-axis current reference $i^*_{d,\text{sh}}$ to the shunt inner loop, ensuring DC link energy balance;
3) **Ancillary service controller** — generates the q-axis reference $i^*_{q,\text{sh}}$ according to the reactive power demand Q^* dispatched by the DSO;
4) **Series voltage controller** — computes the required injection voltage $\boldsymbol{v}^*_{\text{inj}}$ to track $\boldsymbol{v}_{\text{ref}}$ at the load terminals, using a proportional-resonant (PR) regulator in the stationary frame or a PI regulator with feed-forward in the dq frame.

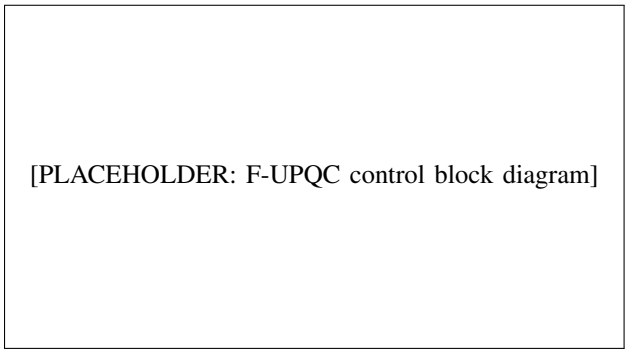


Figure 2: F-UPQC decoupled control architecture.

B. Shunt Inner Current Loop

The current controller cancels the cross-coupling terms $\pm\omega_0 L_{\text{sh}}$ by feed-forward, yielding two independent first-order plants. A proportional-integral (PI) controller

$$C_{\text{sh}}(s) = K_{p,\text{sh}} + \frac{K_{i,\text{sh}}}{s} \tag{8}$$

is designed to achieve a closed-loop bandwidth $\omega_{c,\text{sh}} \approx 2\pi \cdot 1\,\text{kHz}$, well below the switching frequency.

C. DC Bus Voltage Outer Loop

The outer voltage loop bandwidth is set one decade below the inner current loop:

$$C_{\text{dc}}(s) = K_{p,\text{dc}} + \frac{K_{i,\text{dc}}}{s}, \qquad \omega_{c,\text{dc}} \approx 2\pi \cdot 100\,\text{Hz} \tag{9}$$

This separation of timescales ensures that DC bus dynamics do not interfere with the inner current regulation.

D. Series Voltage Compensation Loop

The series controller operates as a voltage feed-forward compensator. The reference injection voltage is computed as:

$$\boldsymbol{v}^*_{\text{inj}} = \boldsymbol{v}_{\text{ref}} - \boldsymbol{v}_{\text{pcc}} + Z_{\text{line}}\, \hat{\boldsymbol{i}}_{\text{load}} \tag{10}$$

where $\hat{\boldsymbol{i}}_{\text{load}}$ is the estimated load current. A PI regulator with bandwidth $\omega_{c,\text{se}}$ closes the tracking loop.

V. Steady-State Analysis and Component Rating

A. Power Flow

Under balanced, steady-state conditions with a purely reactive ancillary service demand $Q^* \neq 0$ and no active series compensation ($\boldsymbol{v}_{\text{inj}} = 0$), the active power balance of the DC bus requires:

$$P_{\text{sh}} + P_{\text{se}} + P_{\text{sto}} = P_{\text{loss}} \tag{11}$$

Since reactive power exchange does not transfer net active energy, $P_{\text{sh}} \approx P_{\text{loss}}$, and the storage is idle ($P_{\text{sto}} \approx 0$) during normal operation.

During a voltage sag of depth ΔV and duration τ, the series VSC must inject active power:

$$P_{\text{se,sag}} = \frac{\Delta V}{V_{\text{nom}}}\, P_{\text{load}} \tag{12}$$

If the shunt VSC cannot fully supply this active power from the grid (e.g., during a deep or long sag), the energy deficit

$$E_{\text{sto}} = P_{\text{se,sag}}\, \tau \tag{13}$$

must be supplied by the storage element, which thus determines the minimum storage capacity for a given sag profile.

B. Converter Rating

Table I summarises the nominal ratings of the main power components for a representative 100 kVA installation.

Table I: Nominal Component Ratings (Representative 100 kVA System)

Component	Parameter	Value
Shunt VSC	Apparent power	S_{sh} [kVA]
	DC link voltage	V_{dc} [V]
	Coupling inductance	L_{sh} [mH]
Series VSC	Apparent power	S_{se} [kVA]
	Injection transformer ratio	n_s
	Series inductance	L_{se} [mH]
DC bus	Capacitance	C_{dc} [mF]
Storage	Energy capacity	E_{sto} [kWh]
	Peak power	P_{sto} [kW]

VI. Regulatory Alignment

The F-UPQC topology is well-positioned within the evolving European regulatory framework for active distribution networks.

The **EU Clean Energy Package** [8] mandates that DSOs transition from passive network managers to active facilitators of distributed flexibility. DSO Entity, representing over 900 distribution operators across 27 EU member states, is developing flexibility assessment methodologies and market frameworks that require grid-connected devices to be capable of providing reactive power support and harmonic mitigation on demand [5].

The **EN 50160** standard defines voltage quality characteristics at the point of supply, establishing the tolerance bands for voltage magnitude, harmonic distortion, and unbalance that the series compensation loop of the F-UPQC is designed to enforce at the load terminals.

The **IEC 61000-3-x** series governs harmonic current emission limits for equipment connected to public networks; the shunt converter of the F-UPQC can be programmed to ensure compliance of the load seen from the PCC with these limits, effectively acting as an active front end (AFE) for the entire facility.

Finally, the **EU Network Code on Demand Response** (currently under development alongside Implementing Regulation on data requirements for demand response [8]) establishes the conditions under which grid-connected inverters can participate in distribution-level balancing and flexibility markets. The F-UPQC, with its bidirectional DC bus and storage interface, is a natural candidate for participation in such markets, offering a revenue stream that can partially offset capital expenditure.

VII. Conclusion

This paper has presented the Flipped UPQC (F-UPQC), a novel topology for medium-voltage and low-voltage active distribution networks that simultaneously addresses two emerging requirements: the protection of sensitive industrial loads from voltage disturbances, and the provision of ancillary services to the distribution system operator.

The key innovation of the F-UPQC with respect to the classical UPQC lies in the topological inversion of the two converters: the shunt VSC faces the grid, enabling reactive power injection and harmonic mitigation at the PCC, while the series VSC faces the load, ensuring clean voltage delivery regardless of grid disturbances. An optional energy storage element on the shared DC bus extends compensation capability to sustained sag events and enables participation in DSO flexibility markets.

A complete mathematical model in the synchronous dq reference frame has been derived, and a decoupled four-loop control architecture has been proposed. Steady-state analysis confirms the feasibility of simultaneous dual-mode operation and provides sizing guidelines for the main power components.

Future work will focus on:

- Hardware-in-the-loop (HIL) validation using a dSPACE real-time platform;
- Optimisation of storage sizing for participation in distribution-level flexibility markets under EN 50160 compliance constraints;
- Extension of the topology to three-phase four-wire systems to address voltage unbalance in LV grids.

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